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NMAM INSTITUTE OF TECHNOLOGY, NITTE**Off-Campus Centre of Nitte (Deemed to be University)****Third Semester B.Tech (AI&DS/AI&ML/CCE/CSE/ISE) (CBCS) Degree Examinations**

December 2023

CS2002-1 – OBJECT ORIENTED PROGRAMMING

(For AI&DS, AI&ML, CCE, CSE, ISE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **Which statement is true about Java?**
A) Java is a sequence-dependent programming language
B) Java is a code dependent programming language
C) Java is a platform-dependent programming language
D) Java is a platform-independent programming language
2. **What is not the use of “this” keyword in Java?**
A) Referring to the instance variable when a local variable has the same name
B) Passing itself to the method of the same class
C) Passing itself to another method
D) Calling another constructor in constructor chaining
3. **Which of the following is a superclass of every class in Java?**
A) ArrayList
B) Abstract class
C) Object class
D) String
4. **Which of these packages contains the exception Stack Overflow in Java?**
A) java.io
B) java.system
C) java.lang
D) java.util
5. **Which of these keywords are used for the block to be examined for exceptions?**
A) check
B) throw
C) catch
D) try
6. **Which one of the following is not an access modifier?**
A) Protected
B) Void
C) Public
D) Private
7. **What is false about constructor?**
A) Constructors cannot be synchronized in Java
B) Java does not provide default copy constructor
C) Constructor can have a return type
D) “this” and “super” can be used in a constructor
8. **Which of the following is a valid declaration of an object of class Box?**
A) Box obj = new Box();
B) Box obj = new Box;
C) obj = new Box();
D) new Box obj;
9. **What is the process of defining more than one method in a class differentiated by method signature?**
A) Function overriding
B) Function overloading
C) Function doubling
D) Function call
10. **Which of the following is a method having same name as that of it's class?**
A) finalize
B) delete
C) class
D) constructor

11. Which of this statement is incorrect?
 A) All object of a class are allotted memory for the all the variables defined in the class
 B) If a function is defined public it can be accessed by object of other class by inheritance
 C) main() method must be made public
 D) All object of a class are allotted memory for the methods defined in the class
12. Which of these is correct about passing an argument by call-by-value process?
 A) Copy of argument is made into the formal parameter of the subroutine
 B) Reference to original argument is passed to formal parameter of the subroutine
 C) Copy of argument is made into the formal parameter of the subroutine and changes made on parameters of subroutine have effect on original argument
 D) Reference to original argument is passed to formal parameter of the subroutine and changes made on parameters of subroutine have effect on original argument
13. What is the process of defining a method in terms of itself, that is a method that calls itself?
 A) Polymorphism
 B) Recursion
 C) Encapsulation
 D) Abstraction
14. Which of these keywords can be used to prevent inheritance of a class?
 A) super
 B) constant
 C) class
 D) final
15. Which of these class relies upon its subclasses for complete implementation of its methods?
 A) Object class
 B) abstract class
 C) ArrayList class
 D) Base class
16. Which of these is correct way of inheriting class A by class B?
 A) class B + class A {}
 B)) class B inherits class A {}
 C) class B extends A {}
 D) class B extends class A {}
17. Which of these exceptions handles the situations when an illegal argument is used to invoke a method?
 A) IllegalArgumentException
 B) Argument Exception
 C) IllegalArgumentExpection
 D) IllegalMethodArgumentExpection
18. Which of these class is used to create user defined exception?
 A) java.lang
 B) Exception
 C) RunTime
 D) System
19. When is the finalize() method called?
 A) Before Garbage Collection
 B) Before an object goes out of scope
 C) Before a variable goes out of scope
 D) Before the Main method call
20. What is Runnable?
 A) Abstract class
 B) Interface
 C) Class
 D) Method

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain Command line Arguments, Write Java program to take values from user from command line and find the average of these values.	8	L*2	1	1
	b) Create a class that models student information, including student number, branch name, semester, and CGPA. Implement constructors to set initial values, methods to modify the semester, display student details, and update the CGPA.	8	L3	1	2
2.	a) Write a Java program that demonstrates the usage of objects as both method arguments and return types.	8	L2	1	2
	b) Explain the concept of package visibility in Java and provide relevant examples to illustrate it.	8	L2	2	1

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| 3. a) | With an example explain the usage of super keyword in java. | 8 | L2 | 2 | 2 |
| b) | Create a class Account with two overloaded constructors. The first constructor is used for initializing, the name of account holder, the account number and the initial amount in the account. The second constructor is used for initializing the name of the account holder, the account number, the type of account and the current balance. The class Account is having methods Deposit (), Withdraw (), and Get_Balance(). Make the necessary assumption for data members and return types of these methods. Write a Java program to create the objects of Account class and use them. | 8 | L3 | 2 | 2 |

Unit – II

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|-------|---|---|----|---|---|
| 4. a) | Write a program in java to read an integer number from user. An exception should be thrown if number entered is less than 10. Handle the exception in the same program. | 8 | L2 | 2 | 2 |
| b) | Explain need for synchronization with an example program. | 8 | L2 | 3 | 2 |
| 5. a) | What is the difference between 'throw' and 'throws' in Exception Handling? Explain with suitable examples. | 8 | L2 | 2 | 2 |
| b) | Describe the thread life cycle in Java. | 8 | L2 | 3 | 2 |
| 6. a) | Design a java program to illustrate the need of synchronization using synchronized method. | 8 | L2 | 3 | 2 |
| b) | Explain how Java manages exceptions and their handling with suitable pseudocode. | 8 | L2 | 2 | 1 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Explain what is the Collection framework. Describe Methods of Iterator interface. | 8 | L2 | 5 | 2 |
| b) | Write a program to handle any 2 Mouse Event Handling. | 8 | L3 | 4 | 2 |
| 8. a) | Illustrate Java Array List class with suitable examples. | 8 | L2 | 5 | 2 |
| b) | With an example code, explain the event handling in Java for JButton. | 8 | L3 | 4 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
